Homework 7

Question 1

The exact equation for the $\frac{d\bar{P}}{d\Omega}$ is given by

$$\frac{dP}{d\Omega} = \frac{1}{4\pi}\frac{1}{c^3}\left[\hat{r} \times \int d^3 r \, \frac{\partial \vec{j}(r', t_r)}{\partial t}\right]^2 \quad -----(1)$$

where $t_r = t - \frac{r}{c} + \frac{\left(\hat{r}.\vec{r'}\right)}{c}$

Equation 10.233 from the notes was

$$\vec{j} = \begin{pmatrix} 0 \\ 0 \\ j_z \end{pmatrix} \quad \text{here } j_z = I \sin\left(\frac{kl}{2} - k \left| z \right|\right) \delta(x)\, \delta(y) \sin(\omega t)$$

Here $\omega = kc$ and $\left| z \right| \leq \frac{l}{2}$

We can put the above expression in equation (1) and we can get the equation 10.239

$$\frac{dP}{d\Omega} = \frac{1}{(4\pi)^2 c} 4 I^2 \frac{\cos^2\left(\omega\left(t - \frac{r}{c}\right)\right)}{\sin^4(\theta)}\left[\cos\left(\frac{kl}{2}\cos(\theta)\right) - \cos\left(\frac{kl}{2}\right)\right]^2 \sin^2(\theta)$$

We can arrange this like so

Now if $kl << 1$, then we can expand the $\cos(\theta) = 1 - \frac{\theta^2}{2}$

$$\frac{dP}{d\Omega} \sim \frac{I^2 \cos^2\left(\omega\left(t - \frac{r}{c}\right)\right)}{(4\pi)^2 c} \frac{\left[1 - \frac{1}{2}\left(\frac{kl}{2}\cos(\theta)\right)^2 - \left(1 - \frac{1}{2}\left(\frac{kl}{2}\right)^2\right)\right]^2}{\sin^2(\theta)} = \frac{I^2 \cos^2(\omega(t)}{(4\pi)^2 c} \frac{\left(\frac{kl}{2}\right)^4 \left[1 - \cos^2(\theta)\right]^2}{\sin^2(\theta)} = \frac{I^2 \cos^2(\omega(t)}{(4\pi)^2 c}\left(\frac{kl}{2}\right)^4 \sin^2(\theta) \quad \textbf{-- -- (A)}$$

For average we have take the average value of $\cos(\omega t)$ and by which we will come to the result

$$\frac{(d\bar{P})}{d\Omega} = \frac{(I^2)}{2c(4\pi)^2}\left(\frac{kl}{2}\right)^4 \sin^2(\theta)$$

Equation 10.207 Which tell us about the power radiated by diple

$$\frac{dP}{d\Omega} = \frac{1}{(4\pi)^2}\frac{1}{c^3}\ddot{d}^2 \sin^2(\theta) \quad ------(2)$$

Defination of the d is

$$\vec{d} = \int \vec{r}\, \rho\left(\vec{r}, t\right) d^3 r = \text{electric dipole moment}$$

$$\left(\ddot{\vec{d}}\right) = \frac{\partial^2}{\partial t^2}\int \vec{r}\, \rho\left(\vec{r}, t\right) d^3 r = \frac{\partial}{\partial t}\int j\left(\vec{r}, t\right) d^3 r = \frac{\partial}{\partial t}\sin(\omega t)\int_{-\frac{l}{2}}^{\frac{l}{2}} I \sin\left(\frac{kl}{2} - k\left|z\right|\right) dz$$

$$\left(\ddot{\vec{d}}\right) = \left(2\frac{\partial}{\partial t}\sin(\omega t)\, I\left(1 - \cos\left(\frac{kl}{2}\right)\right)\right)^2$$

Simplifying we will get

$$\left(\ddot{\vec{d}}\right)^2 = 16\, c^2\, I^2 \cos^2(\omega t) \sin^4\left(\frac{kl}{4}\right)$$

Putting this result in the (2) we will get

$$\frac{dP}{d\Omega} = \frac{1}{(4\pi)^2}\, \frac{1}{c^3}\, 16\, c^2\, I^2 \cos^2(\omega t) \sin^4\left(\frac{kl}{4}\right) \sin^2(\theta)$$

$$\frac{dP}{d\Omega} = \frac{\left(16\, I^2 \cos^2(\omega t)\right)}{(4\pi)^2\, c} \sin^4\left(\frac{kl}{4}\right) \sin^2(\theta)$$

Now we can expand the sine using $kl << 1$:

$$\frac{dP}{d\Omega} = \frac{\left(I^2 \cos^2(\omega t)\right)}{(4\pi)^2\, c} \left(\frac{kl}{2}\right)^4 \sin^2(\theta) \quad \text{------} \; \boldsymbol{(B)}$$

$$\frac{(\overline{dP})}{d\Omega} = \frac{\left(I^2\, \overline{\cos^2}(\omega t)\right)}{(4\pi)^2\, c} \left(\frac{kl}{2}\right)^4 \sin^2(\theta) = \frac{\left(I^2\right)}{2\, c (4\pi)^2} \left(\frac{kl}{2}\right)^4 \sin^2(\theta)$$

In both case we found that same power distribution . This happen because in the far field radiation limit, angular difference are negligible and dipole term is dominant so the dipole approximation makes *a* good model.

part 2

Power radiated by the half wave antenna

$$\frac{\overline{dP}}{d\Omega} = \frac{\left(2\, I^2\right)}{(4\pi)^2\, c}\, \frac{\cos^2\left(\frac{\pi}{2}\cos(\theta)\right)}{\sin^2(\theta)}$$

Total power radiated is given by

$$\int \frac{\overline{dP}}{d\Omega}\, d\Omega = \int \frac{\left(2\, I^2\right)}{(4\pi)^2\, c}\, \frac{\cos^2\left(\frac{\pi}{2}\cos(\theta)\right)}{\sin^2(\theta)}\, d\Omega$$

$$P_{hw} = \frac{\left(2\, I^2\right)}{(4\pi)^2\, c} \int \frac{\cos^2\left(\frac{\pi}{2}\cos(\theta)\right)}{\sin^2(\theta)}\, d\Omega$$

$$P_{hw} = \frac{\left(2\, I^2\right)}{(4\pi)^2\, c} \int_0^{2\pi} d\phi \int_0^{\pi} \frac{\cos^2\left(\frac{\pi}{2}\cos(\theta)\right)}{\sin^2(\theta)} \sin(\theta)\, d\theta$$

Integrating

$$P_d = 1.22\, \frac{I^2}{4\pi c}$$

For dipole soution

$$P_d = \frac{I^2}{(4\pi)^2\, 2\, c} \left(\frac{kl}{2}\right)^4 2\pi \int_0^{\pi} \sin^3\theta\, d\theta$$

Inetgrating we will get

$$P_d = \frac{\pi^3}{48}\, \frac{I^2}{4\pi c}$$

$$\frac{P_{Hw}}{P_D} = 1.89$$

Hence we can say that half wave' *s* antena power is 1.89 times of the dipole approimation.

Question 2

1)

$$J_z = I\,\sin\left(\frac{kl}{2} - k\,|\,z\,|\right)\delta(y)\sin(\omega t)[\delta(x) + \delta(x-a)]$$

$$\frac{dP}{d\Omega} = \frac{1}{4\pi}\frac{1}{c^3}\left[\hat{r} \times \int d^3 r\,\frac{\partial \vec{j}(r', t_r)}{\partial t}\right]^2$$

$$\frac{\partial \vec{j}}{\partial t} = \frac{\partial}{\partial t}\left(I\,\sin\left(\frac{kl}{2} - k\,|\,z'\,|\right)\delta(y)\sin\left(\omega\left(t - \frac{r}{c} + \frac{(\hat{r}.\vec{r'})}{c}\right)[\delta(x) + \delta(x-a)]\right)\right)$$

$$\frac{\partial \vec{j}}{\partial t} = \left(\omega I\,\sin\left(\frac{kl}{2} - k\,|\,z'\,|\right)\delta(y)\cos\left(\omega\left(t - \frac{r}{c} + \frac{(\hat{r}.\vec{r'})}{c}\right)[\delta(x) + \delta(x-a)]\right)\right)$$

$$\int d^3 r\left[\frac{\partial \vec{j}(r', t_r)}{\partial t}\right] = \int_{\frac{-l}{2}}^{\frac{l}{2}}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\omega I\,\sin\left(\frac{kl}{2} - k\,|\,z'\,|\right)\delta(x)\,\delta(y)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{(\hat{r}.\vec{r'})}{c}\right)\right)\right)dx\,dy\,dz +$$

$$\int_{\frac{-l}{2}}^{\frac{l}{2}}\int_{-\infty}^{\infty}\int_{\infty}^{\infty}\omega I\,\sin\left(\frac{kl}{2} - k\,|\,z'\,|\right)\delta(x-a)\,\delta(y)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{(\hat{r}.\vec{r'})}{c}\right)\right)\right)dx\,dy\,dz$$

$$\int d^3 r\left[\frac{\partial \vec{j}(r', t_r)}{\partial t}\right] = \int_{\frac{-l}{2}}^{\frac{l}{2}}\omega I\,\sin\left(\frac{kl}{2} - k\,|\,z'\,|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{(\hat{r}.\vec{r'})}{c}\right)\right)\right)dz \quad \text{at}\,(x' = 0,\ y' = 0) +$$

$$\int_{\frac{-l}{2}}^{\frac{l}{2}}\omega I\,\sin\left(\frac{kl}{2} - k\,|\,z'\,|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{(\hat{r}.\vec{r'})}{c}\right)\right)\right)dz\ \text{at}\,((x' = a),\ y' = 0)$$

$$\int d^3 r\left[\frac{\partial \vec{j}(r', t_r)}{\partial t}\right] = \int_{\frac{-l}{2}}^{\frac{l}{2}}\omega I\,\sin\left(\frac{kl}{2} - k\,|\,z'\,|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{(\hat{r}.\vec{r'})}{c}\right)\right)\right)dz \quad \text{at}\,(x' = 0,\ y' = 0) +$$

$$\int_{\frac{-l}{2}}^{\frac{l}{2}}\omega I\,\sin\left(\frac{kl}{2} - k\,|\,z'\,|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{(\hat{r}.\vec{r'})}{c}\right)\right)\right)dz\ \text{at}\,((x' = a),\ y' = 0)$$

For the first integral $\hat{r}.\vec{r}' = z'\cos(\theta)$

$$\hat{r}.\vec{r}' = (\sin(\theta)\cos(\phi),\ \sin(\theta)\sin(\phi),\ \cos(\theta))\begin{pmatrix} a \\ 0 \\ z' \end{pmatrix} = a\sin(\theta)\cos(\phi) + z'\cos(\theta)$$

Let us take the first integral

$\Rightarrow$ Hence first integral is =

$$\int_{\frac{-l}{2}}^{\frac{l}{2}}\omega I\,\sin\left(\frac{kl}{2} - k\,|\,z'\,|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{(\hat{r}.\vec{r'})}{c}\right)\right)\right)dz \quad \text{at}\,(x' = 0,\ y' = 0) =$$

$$\int_{\frac{-l}{2}}^{\frac{l}{2}}\omega I\,\sin\left(\frac{kl}{2} - k\,|\,z'\,|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{z'\cos(\theta)}{c}\right)\right)\right)dz$$

$$\Rightarrow \int_{-\frac{l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\,|z'|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{z'\cos(\theta)}{c}\right)\right)\right) dz =$$

$$\int_{-\frac{l}{2}}^{\frac{l}{2}} \omega I \left\{\sin\left(\frac{kl}{2} - k\,|z'|\right)\cos\left(\left(\omega\left(t - \frac{r}{c}\right)\right)\cos\left(\frac{\omega z'\cos(\theta)}{c}\right) - \sin\left(\frac{kl}{2} - k\,|z'|\right)\sin\left(\left(\omega\left(t - \frac{r}{c}\right)\right)\sin\left(\frac{\omega z'\cos(\theta)}{c}\right)\right\}\right.$$

dz

$$\Rightarrow \text{Ist integral} = \int_{-\frac{l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\,|z'|\right)\cos\left(\omega\left(t - \frac{r}{c}\right)\right)\cos(k z'\cos(\theta))\, dz$$

Let us now evaluate the second integral

$$\int_{-\frac{l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\,|z'|\right)\cos\left(\left(\omega\left(t - \frac{r}{c} + \frac{(\hat{r}.\vec{r'})}{c}\right)\right)\right) dz \text{ at } ((x' = a),\ y' = 0) =$$

$$\int_{-\frac{l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\,|z'|\right)\cos\left(\left(\omega\left(t - \frac{r}{c}\right) + k a \sin(\theta)\cos(\phi) + k z'\cos(\theta)\right)\right) dz$$

$$\Rightarrow \text{2 nd Integral} = \int_{-\frac{l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\,|z'|\right)\cos\left(\left(\omega\left(t - \frac{r}{c}\right) + k a \sin(\theta)\cos(\phi)\right) + k z'\cos(\theta)\right) dz'$$

In this integral also sine term will also go away and hence clearing up the clutter

$$\Rightarrow \text{2 nd Integral} = \cos\left(\left(\omega\left(t - \frac{r}{c}\right) + k a \sin(\theta)\cos(\phi)\right)\right)\int_{-\frac{l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\,|z'|\right)\cos(k z'\cos(\theta))$$

Hence combining the 1 st and 2 nd Integral we can

$$\int d^3 r \left[\frac{\partial \vec{j}(r', t_r)}{\partial t}\right] =$$

$$\omega I \hat{z}\left[\cos\left(\omega\left(t - \frac{r}{c}\right)\right) + \cos\left(\left(\omega\left(t - \frac{r}{c}\right) + k a \sin(\theta)\cos(\phi)\right)\right)\right]\int_{-\frac{l}{2}}^{\frac{l}{2}} \omega I \sin\left(\frac{kl}{2} - k\,|z'|\right)\cos(k z'\cos(\theta))$$

Solving this integral we will get

$$\int d^3 r \left[\frac{\partial \vec{j}(r', t_r)}{\partial t}\right] = 2\, I c\, \hat{z}\left[\cos\left(\omega\left(t - \frac{r}{c}\right)\right) + \cos\left(\left(\omega\left(t - \frac{r}{c}\right) + k a \sin(\theta)\cos(\phi)\right)\right)\right]\frac{\left\{\cos\left(\frac{kl}{2}\cos(\theta)\right) - \cos\left(\frac{kl}{2}\right)\right\}}{\sin^2(\theta)}$$

$$\frac{dP}{d\Omega} = \frac{I^2}{4\pi^2 c}\,\frac{\left\{\cos\left(\frac{kl}{2}\cos(\theta)\right) - \cos\left(\frac{kl}{2}\right)\right\}^2}{\sin^2(\theta)}\left[\cos\left(\omega\left(t - \frac{r}{c}\right)\right) + \cos\left(\left(\omega\left(t - \frac{r}{c}\right) + k a \sin(\theta)\cos(\phi)\right)\right)\right]^2$$

Now to calculate the averaged angular power distribution, we need to find

$$\frac{\left(\int_0^{\frac{2\pi}{\omega}}\left[\cos\left(\omega\left(t - \frac{r}{c}\right)\right) + \cos\left(\left(\omega\left(t - \frac{r}{c}\right) + k a \sin(\theta)\cos(\phi)\right)\right)\right]^2 dt\right)}{\frac{2\pi}{\omega}} = 1 + \cos(k a \sin(\theta)\cos(\phi))$$

Finally we find that time averaged angular power distribution is

$$\frac{dP}{d\Omega} = \frac{I^2}{4\pi^2 c}\,\frac{\left\{\cos\left(\frac{kl}{2}\cos(\theta)\right) - \cos\left(\frac{kl}{2}\right)\right\}^2}{\sin^2(\theta)}\,(1 + \cos(k a \sin(\theta)\cos(\phi)))$$

$$\frac{dP}{d\Omega} = \frac{I^2}{2\pi^2 c}\,\frac{\left\{\cos\left(\frac{kl}{2}\cos(\theta)\right) - \cos\left(\frac{kl}{2}\right)\right\}^2}{\sin^2(\theta)}\cos^2\left(\frac{ka}{2}\sin(\theta)\cos(\phi)\right)$$

2)

Assume that both the antenna are half wave. So kl = π

From the above results.

$$\frac{d\bar{P}}{d\Omega} = \frac{I^2}{2\pi^2 c}\frac{\left\{\cos\left(\frac{\pi}{2}\cos(\theta)\right)\right\}^2}{\sin^2(\theta)}\cos^2\left(\frac{\pi a}{2l}\sin(\theta)\cos(\phi)\right)$$

we have to now look for the time avearged angular distribution for this

Let us look when $\phi = \pm\frac{\pi}{2}$

$$\frac{d\bar{P}}{d\Omega} = \frac{I^2}{2\pi^2 c}\frac{\cos^2\left(\frac{\pi}{2}\cos(\phi)\right)}{\sin^2(\theta)} \Rightarrow \text{Constructive interference along the yz plane.}$$

when $\theta = \frac{\pi}{2}$ and $\phi = 0, \pi$

$$\frac{d\bar{P}}{d\Omega} = \frac{I^2}{2\pi^2 c}\frac{\left\{\cos\left(\frac{\pi}{2}\cos(\theta)\right)\right\}^2}{\sin^2(\theta)}\cos^2\left(\frac{\pi a}{2l}\right)$$

Destructive interference will happen when $a\sin(\theta)\cos(\phi) = l$

as $\frac{d\bar{P}}{d\Omega} = \frac{I^2}{2\pi^2 c}\frac{\left\{\cos\left(\frac{\pi}{2}\cos(\theta)\right)\right\}^2}{\sin^2(\theta)}\cos^2\left(\frac{\pi}{2}\right) = 0$

Question 3

Power radiated by particle of charge ze and mass m passes through a force field $\vec{F}$

$$P = \frac{((ze)^2 \gamma^2)}{6\pi m^2 c^3}\left[F^2 - \left(\vec{\beta}.\vec{F}\right)^2\right] \text{ where } \vec{\beta} = \frac{\vec{v}}{c}$$

From equation 13.11 we can say that

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^3\left[(\dot{v})^2 - \frac{1}{c^2}(v \times \dot{v})^2\right]$$

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^3\left[(a)^2 - (\boldsymbol{\beta} \times \boldsymbol{a})^2\right]$$

We have to now to express the accelration in terms of the force

$$F^\mu = m\frac{\mathrm{du}^\mu}{\mathrm{d}\tau} = m\frac{d}{\mathrm{dt}}\gamma\begin{pmatrix} c \\ v^i \end{pmatrix} = \mathrm{m}\gamma\left[\frac{\left(\vec{v}.\vec{a}\right)}{c^2}\gamma^3\begin{pmatrix} c \\ v^i \end{pmatrix} + \gamma\begin{pmatrix} 0 \\ a^i \end{pmatrix}\right]$$

Extracting 3 Vector of the force is

$$\vec{F} = \gamma^4\frac{m}{c^2}\left(\vec{v}.\vec{a}\right)\vec{v} + \mathrm{m}\gamma^2\vec{a}$$

We have to decompose the F in the parallel and perpendicular components

$$\vec{F} = \gamma^4\frac{m}{c^2}\left(\vec{v}.(a_{\|} + a_{\perp})\right)\vec{v} + \mathrm{m}\gamma^2(a_{\|} + a_{\perp})$$

$$F_{\|} + F_{\perp} = \gamma^4\frac{m}{c^2}(a_{\|})v^2 + \mathrm{m}\gamma^2(a_{\|} + a_{\perp})$$

$$F_{\|} + F_{\perp} = \gamma^4\frac{m}{c^2}(a_{\|})v^2 + \mathrm{m}\gamma^2(a_{\|}) + \mathrm{m}\gamma^2(a_{\perp})$$

$$F_{\|} = \left(\gamma^4\frac{m}{c^2}v^2 + \mathrm{m}\gamma^2\right)a_{\|} \quad \text{and } F_{\perp} = \mathrm{m}\gamma^2(a_{\perp})$$

Let us solve for the $a_{\|}$

$$F_{\|} = \mathrm{m}\gamma^2\left(\gamma^2\frac{1}{c^2}v^2 + 1\right)a_{\|} \qquad \text{writing} \quad \gamma^2\frac{1}{c^2}v^2 + 1 = \frac{1}{\left(1-\frac{v^2}{c^2}\right)}\frac{v^2}{c^2} + 1 = \frac{v^2}{c^2-v^2} + 1 = \frac{c^2}{c^2-v^2} = \gamma^2$$

$$F_{\|} = \mathrm{m}\gamma^4 a_{\|} \quad \Rightarrow a_{\|} = \frac{F_{\|}}{\mathrm{m}\gamma^4} \text{ and } a_{\perp} = \frac{F_{\perp}}{\mathrm{m}\gamma^2}$$

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^6\left[(a_{\|})^2 + (a_{\perp})^2 - (\boldsymbol{\beta} \times (\boldsymbol{a}_{\|} + a_{\perp}))^2\right]$$

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^6\left[(a_{\|})^2 + (a_{\perp})^2 - (\boldsymbol{\beta}\,\boldsymbol{a}_{\perp})^2\right]$$

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^6\left[(a_{\|})^2 + \frac{(a_{\perp})^2}{\gamma^2}\right]$$

Putting the above expressions

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^6\left[\left(\frac{F_{\|}}{\mathrm{m}\gamma^4}\right)^2 + \frac{1}{\gamma^2}\left(\frac{F_{\perp}}{\mathrm{m}\gamma^2}\right)^2\right]$$

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi c^3}\gamma^6\left[\frac{F_{\|}}{m^2\gamma^8} + \frac{F_{\perp}}{m^2\gamma^6}\right]$$

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi m^2 c^3}\left[\frac{F_{\|}}{\gamma^2} + F_{\perp}\right]$$

$F^2 - \left(\vec{\beta}.\vec{F}\right)^2 = (F_{\|})^2 + (F_{\perp})^2 - \beta^2 {F_{\|}}^2 = \frac{{F_{\|}}^2}{\gamma^2} + {F_{\perp}}^2$, as well as the given charge $q = ze$ gives

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi m^2 c^3}\left[F^2 - \left(\vec{\beta}.\vec{F}\right)^2\right]$$

Question 4

We have to show that relativistic result for the instantaneous energy radiated per unit time by the particle of charge ze in $\vec{E}$ and $\vec{B}$ is given by

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi m^2 c^3}\left[\left(\vec{E} + \frac{\vec{v}}{c}\times\vec{B}\right)^2 - \left(\frac{\vec{v}}{c}.\vec{E}\right)^2\right]$$

We have found from the previous result that power emitted by the charge partice in the force field is

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi m^2 c^3}\left[F^2 - \left(\vec{\beta}.\vec{F}\right)^2\right] \qquad ---------- (A)$$

Totale electromagnetic force on the particle is given by

$$\vec{F} = ze\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right)$$

Let us calculate the

$$F^2 = ze\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right).ze\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right) = (ze)^2\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right)^2$$

let us evaluate now

$$\vec{\beta}.\vec{F} = ze\,\vec{\beta}.\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right) = ze\left(\vec{\beta}.\vec{E} + \vec{\beta}.\frac{\vec{v}\times\vec{B}}{c}\right) = ze\left(\vec{\beta}.\vec{E}\right)$$

$$\vec{\beta}.\vec{F} = ze\left(\frac{\vec{v}}{c}.\vec{E}\right)$$

Putting everything in the (A) we will get

$$P = \frac{2}{3}\frac{(ze)^2}{4\pi m^2 c^3}\left[(ze)^2\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right)^2 - \left(ze\left(\frac{\vec{v}}{c}.\vec{E}\right)\right)^2\right]$$

$$P = \frac{2}{3}\frac{(ze)^4}{4\pi m^2 c^3}\left[\left(\vec{E} + \frac{\vec{v}\times\vec{B}}{c}\right)^2 - \left(\frac{\vec{v}}{c}.\vec{E}\right)^2\right]$$

Question 5

a

Energy radiated into the solid angle dΩ per unit retarded time is given by

Equation 15.25 for the 1 particle is written by

$$\frac{dP(t')}{d\Omega} = \frac{\frac{e^2}{(4\pi)^2 c^3}\left[\hat{r}\times\left(\vec{a}+\hat{r}\times\left(\frac{\vec{v}}{c}\times\vec{a}\right)\right)\right]^2}{\left(1-\frac{\vec{v}}{c}.\hat{r}\right)^5} \text{ ---- } (B)$$

$$z(t') = a\cos(\omega_o t')$$

$$\dot{z}(t') = v_z(t') = -a\omega_o \sin(\omega_o t')$$

$$\ddot{z}(t') = a_z(t') = -a(\omega_o)^2\cos(\omega_o t')$$

$$\dot{z}(t') \times \ddot{z}(t') = \vec{v}\times\vec{a} = 0$$

Hence the equation B can be writtten as a

$$\frac{dP(t')}{d\Omega} = \frac{e^2\left[\hat{r}\times\vec{a}\right]^2}{(4\pi)^2 c^3\left(1-\frac{\vec{v}}{c}.\hat{r}\right)^5} \text{ ---------- } C$$

let us calculate the $\hat{r}\times\vec{a}$ and $\vec{v}.\hat{r}$ for our case

$$\hat{r}\times\vec{a} = a(\omega_o)^2\cos(\omega_o t')\left(\sin(\theta)\cos(\phi)\,\hat{x} + \sin(\theta)\sin(\phi)\,\hat{y} + \cos(\theta)\,\hat{z}\right)\times\hat{z}$$

$$\hat{r}\times\vec{a} = a(\omega_o)^2\cos(\omega_o t')\left(-\sin(\theta)\cos(\phi)\,\hat{y} + \sin(\theta)\sin(\phi)\,\hat{x}\right)$$

$$\left(\hat{r}\times\vec{a}\right)^2 = a^2(\omega_o)^4\cos^2(\omega_o t')\,\sin^2(\theta)$$

$$1-\frac{\vec{v}}{c}.\hat{r} = 1 + \frac{1}{c}\left(a\omega_o\sin(\omega_o t')\,\hat{z}\,.\left(\sin(\theta)\cos(\phi)\,\hat{x} + \sin(\theta)\sin(\phi)\,\hat{y} + \cos(\theta)\,\hat{z}\right)\right)$$

$$1-\frac{\vec{v}}{c}.\hat{r} = 1 + \frac{1}{c}\left(a\omega_o\cos(\theta)\sin(\omega_o t')\right) = 1 + \beta\cos(\theta)\sin(\omega_o t')$$

Putting everyting on the equation C

$$\frac{dP(t')}{d\Omega} = \frac{e^2\left[\hat{r}\times\vec{a}\right]^2}{(4\pi)^2 c^3\left(1-\frac{\vec{v}}{c}.\hat{r}\right)^5} = \frac{e^2\left(a^2(\omega_o)^4\cos^2(\omega_o t')\,\sin^2(\theta)\right)}{(4\pi)^2 c^3(1+\beta\cos(\theta)\sin(\omega_o t'))^5}$$

which can be further written as a

$$\frac{dP(t')}{d\Omega} = \frac{e^2 c\,\beta^4\cos^2(\omega_o t')\,\sin^2(\theta)\,)}{(4\pi)^2 a^2(1+\beta\cos(\theta)\sin(\omega_o t'))^5}$$

b

$$\frac{\overline{dP}(t')}{d\Omega} = \int\frac{dP(t')}{d\Omega}\,dt' = \frac{\left(\int_0^{\frac{(2\pi)}{\omega}}\frac{e^2 c\,\beta^4\cos^2(\omega_o t')\,\sin^2(\theta)}{(4\pi)^2 a^2(1+\beta\cos(\theta)\sin(\omega_o t'))^5}\,dt'\right)}{\frac{2\pi}{\omega_0}}$$

$$\frac{\overline{dP}(t')}{d\Omega} = \frac{\omega_0\left(e^2 c\,\beta^4\sin^2(\theta)\right)}{2\pi(4\pi)^2 a^2}\int_0^{\frac{(2\pi)}{\omega_0}}\frac{\cos^2(\omega_o t')\;)}{(1+\beta\cos(\theta)\sin(\omega_o t'))^5}\,dt'$$

Calcuating the inside the integration in the mathematica

$$\frac{\overline{dP}(t')}{d\Omega} = \frac{e^2 c\,\beta^4\sin^2(\theta)}{128\pi)^2 a^2}\,\frac{(4+\beta^2\cos^2(\theta))}{(1-\beta^2\cos^2(\theta))^{\frac{7}{2}}}$$

Question 6

We want to calculate and sketch the frequency distribution for a particle which uniformly decelerates over some time interval τ

accelration $a = \left\{ \begin{array}{ll} \frac{(v_f - v_i)}{\tau}, & \frac{-\tau}{2} \leq t \leq \frac{\tau}{2} \\ 0, & \text{otherwise} \end{array} \right\}$

Let us fourier transform of the accelaration

$$\vec{a}(\omega) = \frac{\hat{2}}{2\pi} \int_{\frac{-\tau}{2}}^{\frac{\tau}{2}} \frac{(v_f - v_i)}{\tau} e^{i\omega t} dt = \frac{1}{i\omega} \frac{(v_f - v_i)}{2\pi\tau} \left(e^{\frac{i\omega\tau}{2}} - e^{-\frac{i\omega\tau}{2}} \right) = \frac{1}{\omega} \frac{(v_f - v_i)}{\pi\tau} \sin\left(\frac{\omega\tau}{2}\right)$$

Energy distribution per unit solid angle and frequncy is given by

$$\frac{dE}{d\Omega d\omega} = \frac{1}{(4\pi)^2} \frac{e^2}{\pi c^3} \left[\hat{r} \times \vec{a}(\omega) \right]^2$$

$$\frac{dE}{d\Omega d\omega} = \frac{1}{(4\pi)^2} \frac{e^2}{\pi c^3} (a(\omega))^2 \sin^2(\theta)$$

$$\frac{dE}{d\omega} = \frac{1}{(4\pi)^2} \frac{e^2}{\pi c^3} (a(\omega))^2 \int d\Omega \sin^2(\theta)$$

putting this into the mathematica we will get

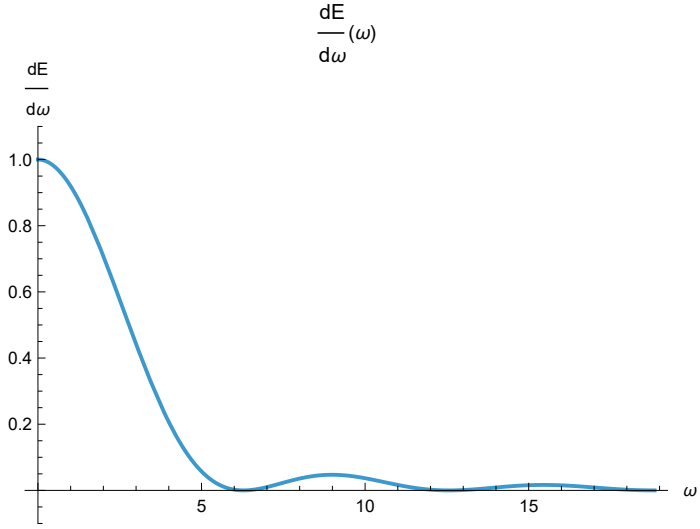
$$\frac{dE}{d\omega} = \frac{e^2}{6\pi^4 c^3} \left(\frac{(v_f - v_i)}{\tau} \right)^2 \frac{\sin^2\left(\frac{\omega\tau}{2}\right)}{\omega^2}$$

$$\frac{dE}{d\omega} = \frac{e^2}{24\pi^4 c^3} ((v_f - v_i))^2 \frac{\sin^2\left(\frac{\omega\tau}{2}\right)}{\left(\frac{\omega\tau}{2}\right)^2}$$

variation of $\frac{dE}{d\omega}$ will be like

In[5]:=
```
Plot[Sin[x/2]^2/(x/2)^2, {x, 0, 6 π}, PlotRange → {-0.1, 1.1},
 PlotLabel → Style["dE/dω (ω)"], AxesLabel → {"ω", "dE/dω"}]
```

Out[5]=



Question 7

A charge e of mass m is going through electric field $(E_o, 0, 0)$ with velocity $(v_0, 0, 0)$ for a distcane x_0
Accelrationn experienced by the charged particle is

$a(t) = \frac{eE_0}{m}$ for $0 < t < t_f$

let us find the fourier transform of the accelaration

$\tilde{a}(\omega) = \frac{1}{2\pi}\int_0^{t_f} a(t)\, e^{i\omega t}\, dt\, \omega$

$\tilde{a}(\omega) = \frac{eE_0}{2\pi i m\omega}\left(e^{i\omega t_f} - 1\right)$

$\tilde{a}(\omega) = \frac{(e E_0)}{\pi m\omega}\sin\left(\frac{\omega t_f}{2}\right) e^{\frac{i\omega t_f}{2}}$

let us now find out the time using the kinematic equation of motion

$x = ut - \frac{1}{2} a t^2$

$\frac{mx_o}{eE_o} = \frac{mv_o}{eE_o} t_f - \frac{1}{2}\frac{eE_0}{m}(t_f)^2$

$\frac{mx_o}{eE_o} + \frac{m^2(v_o)^2}{4e^2(E_o)^2} = \left(t_f + \frac{mv_0}{2eE_o}\right)^2$

hence $t_f = \sqrt{\frac{m}{eE_o}\left(x_o + \frac{mv_o}{4eE_o}\right)} - \frac{mv_o}{2eE_o}$

$\frac{dE}{d\Omega d\omega} = \frac{1}{(4\pi)^2}\frac{e^2}{\pi c^3}(a(\omega))^2 \sin^2(\theta)$

$\frac{dE}{d\omega} = \frac{1}{(4\pi)^2}\frac{e^2}{\pi c^3}(a(\omega))^2 \int d\Omega \sin^2(\theta)$

Putting the value of $a(\omega)$ and calcuting the integral again like in previous question we will get

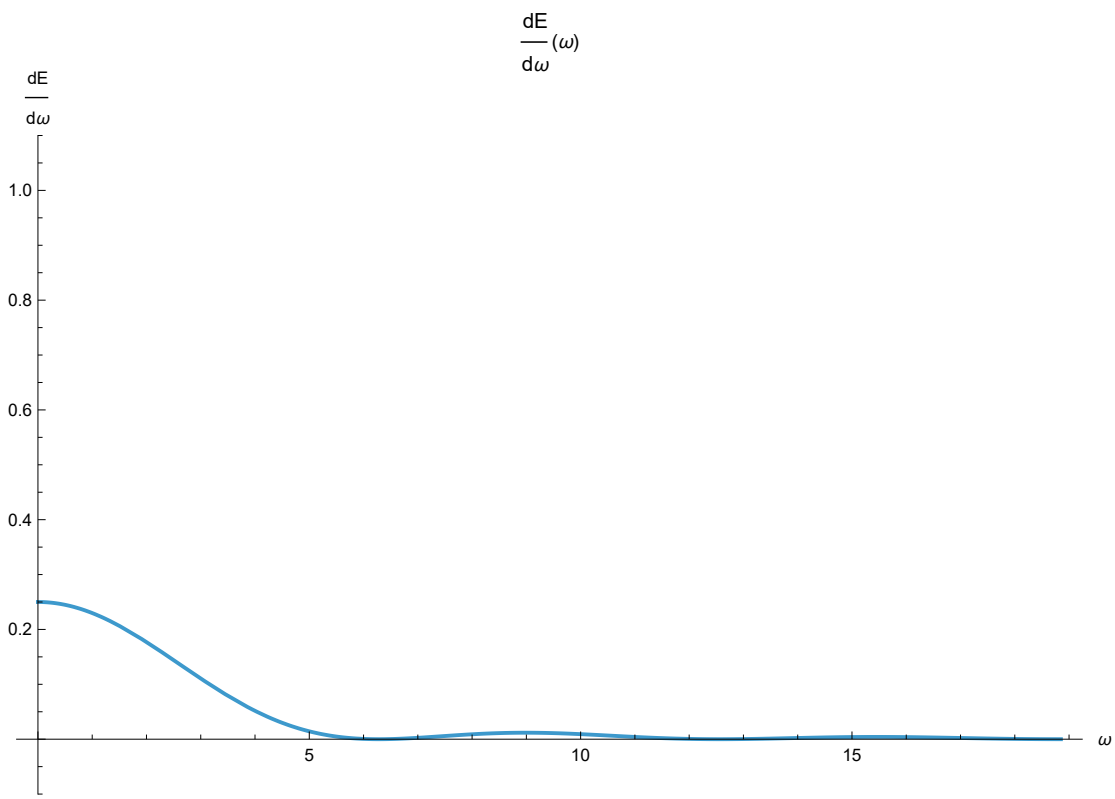
$\frac{dE}{d\omega} = \left(\frac{(e^2 E_o)}{4\pi^2 m c}\right)^2 \frac{2}{c\omega^2}\sin^2\left(\frac{\omega t_f}{2}\right)\frac{4}{3}$

$\frac{dE}{d\omega} = \frac{e^4 (t_f E_o)^2}{6\pi^4 m c^3}\frac{\sin^2\left(\frac{\omega t_f}{2}\right)}{(t_f\omega)^2}\frac{4}{3}$

Plot will be almost same like above

In[8]:=
```
Plot[Sin[x/2]^2/(x)^2, {x, 0, 6 π}, PlotRange → {-0.1, 1.1},
 PlotLabel → Style["dE/dω (ω)"], AxesLabel → {"ω", "dE/dω"}, ImageSize → Large]
```

Out[8]=



Question 8

Given

v(t) = $\vec{v}_o$ = constant

$\dot{v}(t) = 0$

$\vec{v}(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \vec{v}_o\, e^{i\omega t}\, dt = \vec{v}_o\, \delta(\omega)$

$\vec{a}(\omega) = 0$

Plugging into the expression that are given in the questions

$$\frac{dE}{d\Omega d\omega} = \frac{1}{(4\pi)^2}\frac{e^2\omega^2}{\pi c^3}\left[\hat{r}\times\vec{v}(\omega)\right]^2 = \frac{1}{(4\pi)^2}\frac{e^2\omega^2}{\pi c^3}\left[\hat{r}\times\vec{v}_o\,\delta(\omega)\right]^2 = \frac{1}{(4\pi)^2}\frac{e^2\omega^2}{\pi c^3}(v_o)^2\sin^2(\theta)\,\delta^2(\omega)$$

$$\frac{dE}{d\Omega d\omega} = \frac{1}{(4\pi)^2}\frac{e^2}{\pi c^3}\left[\hat{r}\times\vec{a}(\omega)\right]^2 = 0$$

Question 9

Photon density per unit lenght

$\frac{dN}{dz} = \frac{1}{137\,c} \int_{\omega_2}^{\omega_1} 1 - \cos^2(\theta_c)\, d\omega$ ------**D**

where $\text{Cos}(\theta_c) = \frac{1}{\beta n} = \frac{c}{vn}$

we have to find the θ_c , ω_1, ω_2

$\lambda_1 = 4000\,\dot{A}$, $\omega_1 = \frac{(2\pi c)}{\lambda_1} = 4.712 \times 10^{15}$

$\lambda_2 = 6000\,\dot{A}$, $\omega_1 = \frac{(2\pi c)}{\lambda_1} = 3.142 \times 10^{15}$

refractive index = 1.5

KE (Proton) = 10^6 eV = $1.602 \times 10^{-13}\,J$

KE (Proton) = 10^6 eV = $1.602 \times 10^{-10}\,J$

$m(p) = 1.6 \times 10^{-27}$ kg

$m(e) = 9.109 \times 10^{-31}$ kg

$v(e) = \sqrt{2\,\frac{k}{m}} = 4.736 \times 10^8\, m/s$

$v(p) = \sqrt{2\,\frac{k}{m}} = 5.931 \times 10^8\, m/s$

$\theta_c(e) = 70.2$ degree $\theta_c(p) = 62.8$ degree

Putting everything into D and finding

$\frac{dN}{dz}$ for the proton = $30\,236\, m^{-1} \sim 302.36\ \text{cm}^{-1}$

$\frac{dN}{dz}$ for the electron = $33\,873\, m^{-1} \sim 338\ \text{cm}^{-1}$

Hence cherenkov angle for the electron and proton are 70.2 and 62.8 degree respectively. And number of photon emitted are 339 and 302 respectively.